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①

$$\Rightarrow X_0 \Rightarrow \underline{1} e_1 = nC_{1,0} \rightarrow \text{falso}$$

$\Rightarrow Z_n \Rightarrow$ liro do efetivador

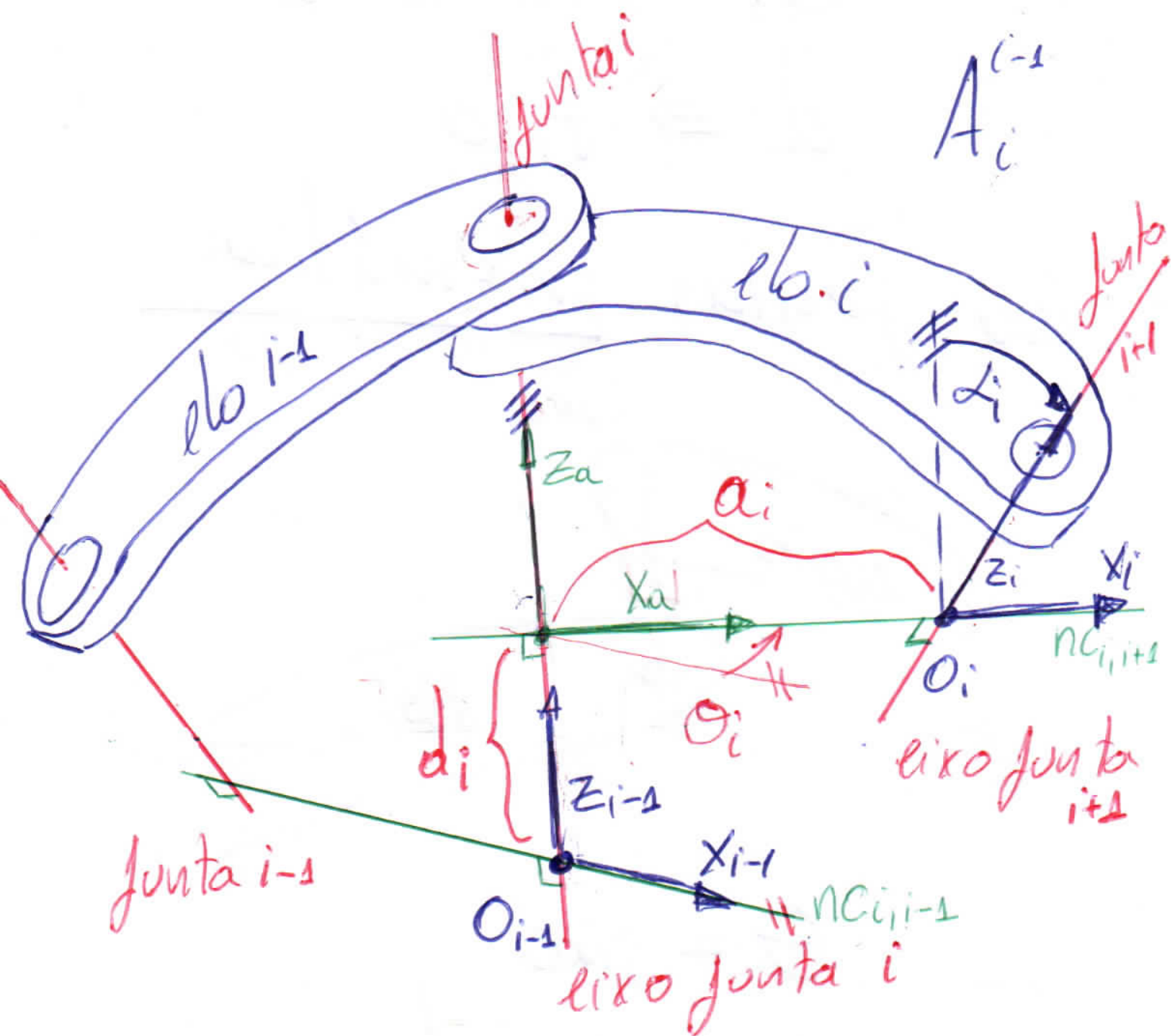
$$X_n \Rightarrow nC_{n,ef}$$

$$Z_{i-1} \Rightarrow C_i$$

2ª etapa

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2º etapa, Identificar os
Parâmetros de DH



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$$A_a^{i-1} = \left[\begin{array}{ccc|c} c_{\theta_i} & -s_{\theta_i} & 0 & 0 \\ s_{\theta_i} & c_{\theta_i} & 0 & 0 \\ \hline 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{array} \right] \quad H/H2$$

$$\hookrightarrow \hat{f}(d_i, \theta_i)$$

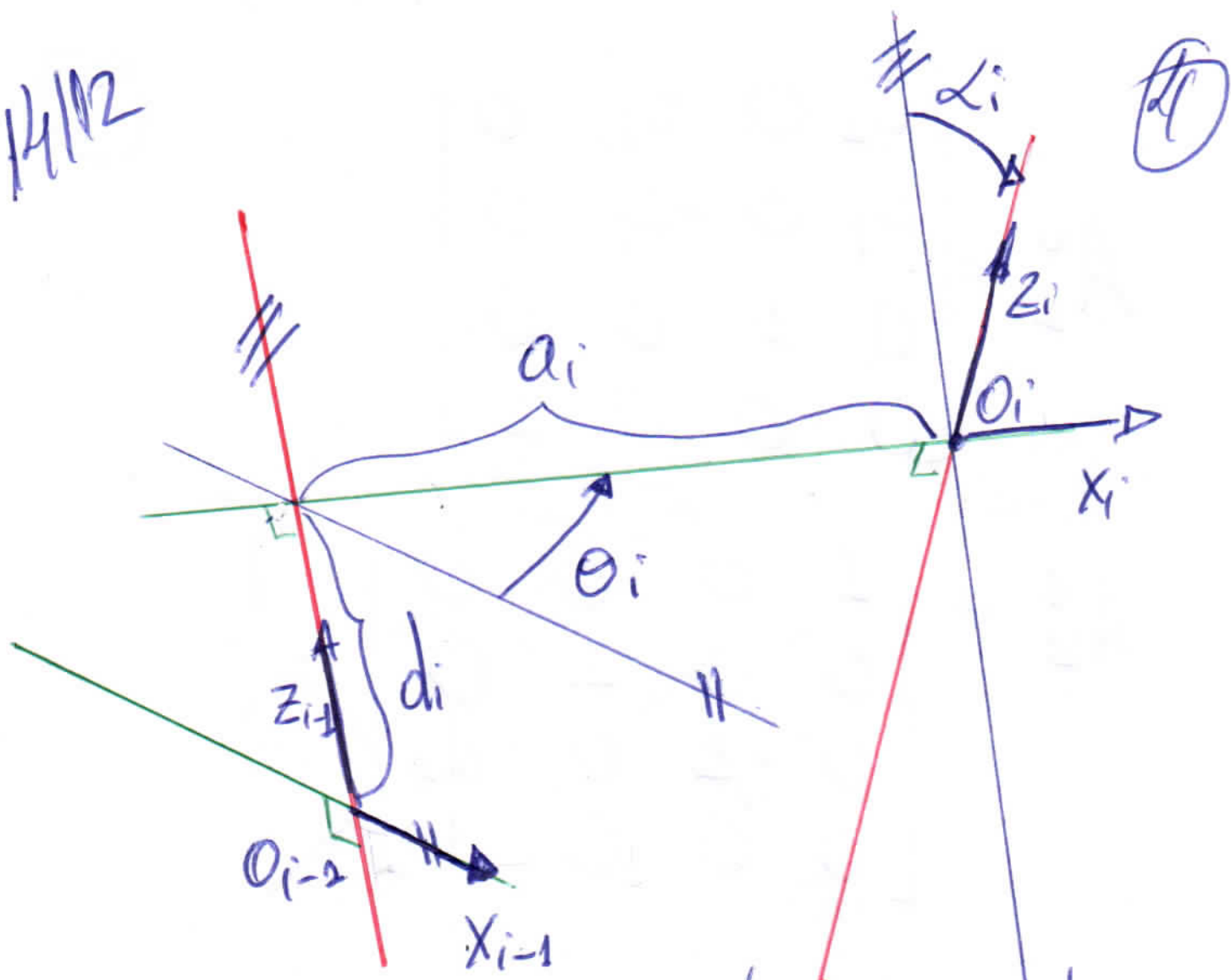
$$A_i^a = \left[\begin{array}{ccc|c} 1 & 0 & 0 & a_i \\ 0 & c_{\alpha_i} & -s_{\alpha_i} & 0 \\ 0 & s_{\alpha_i} & c_{\alpha_i} & 0 \\ \hline 0 & 0 & 0 & 1 \end{array} \right]$$

$$\hookrightarrow f(a_i, \alpha_i)$$

$$A_i^{i-1} = A_a^{i-1} A_i^a$$

$$= \left[\begin{array}{ccc|c} c_{\theta_i} & -s_{\theta_i} c_{\alpha_i} & s_{\theta_i} s_{\alpha_i} & a_i c_{\theta_i} \\ s_{\theta_i} & c_{\theta_i} c_{\alpha_i} & -c_{\theta_i} s_{\alpha_i} & a_i s_{\theta_i} \\ \hline 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{array} \right]$$

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$a_i \Rightarrow$ comprimento do segmento
da normal comum entre os
eixos das junta i e $i+1$

$d_i \Rightarrow$ coordenada da origem O_i em
relação ao sistema ~~anterior~~ na
direção Z_{i-1} $O_{i-1} - X_{i-1} Y_{i-1} Z_{i-1}$

$\alpha_i \Rightarrow$ ângulo entre $Z_{i-1} \xrightarrow{X_i} Z_i$

$\theta_i \Rightarrow$ ângulo entre $X_{i-1} \xrightarrow{Z_{i-1}} X_i$

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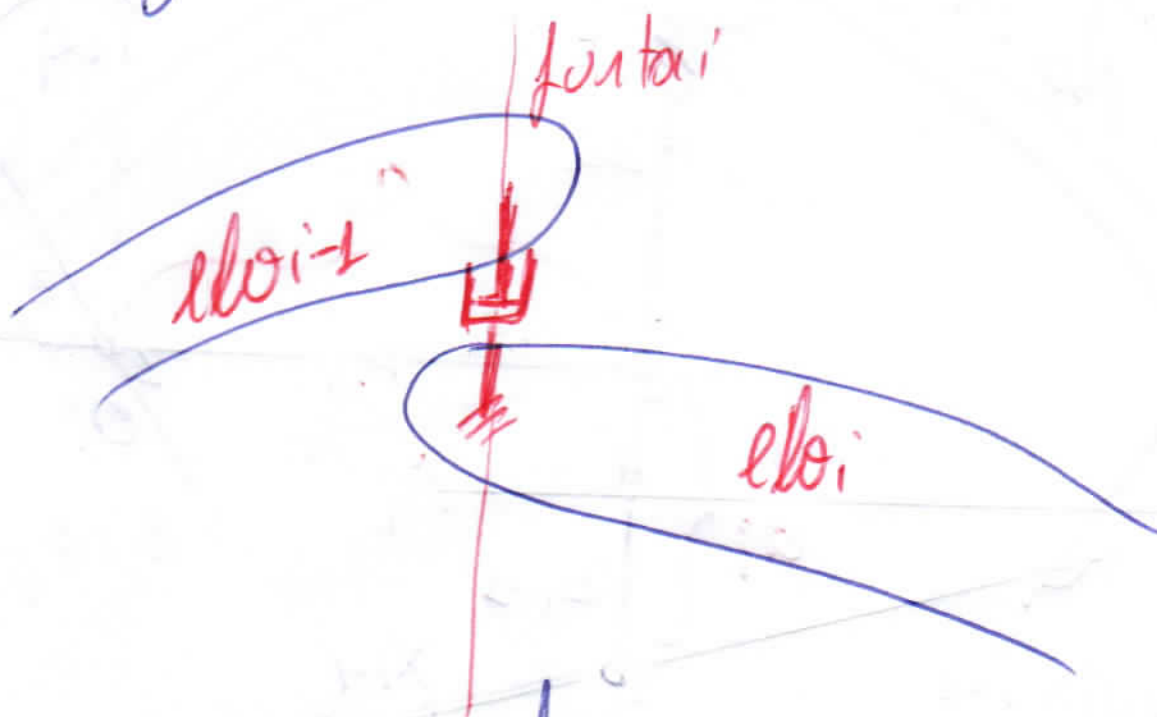
⑤

Para juntas de revolução

$\theta_i \Rightarrow$ variável

$d_i \Rightarrow$ fixo

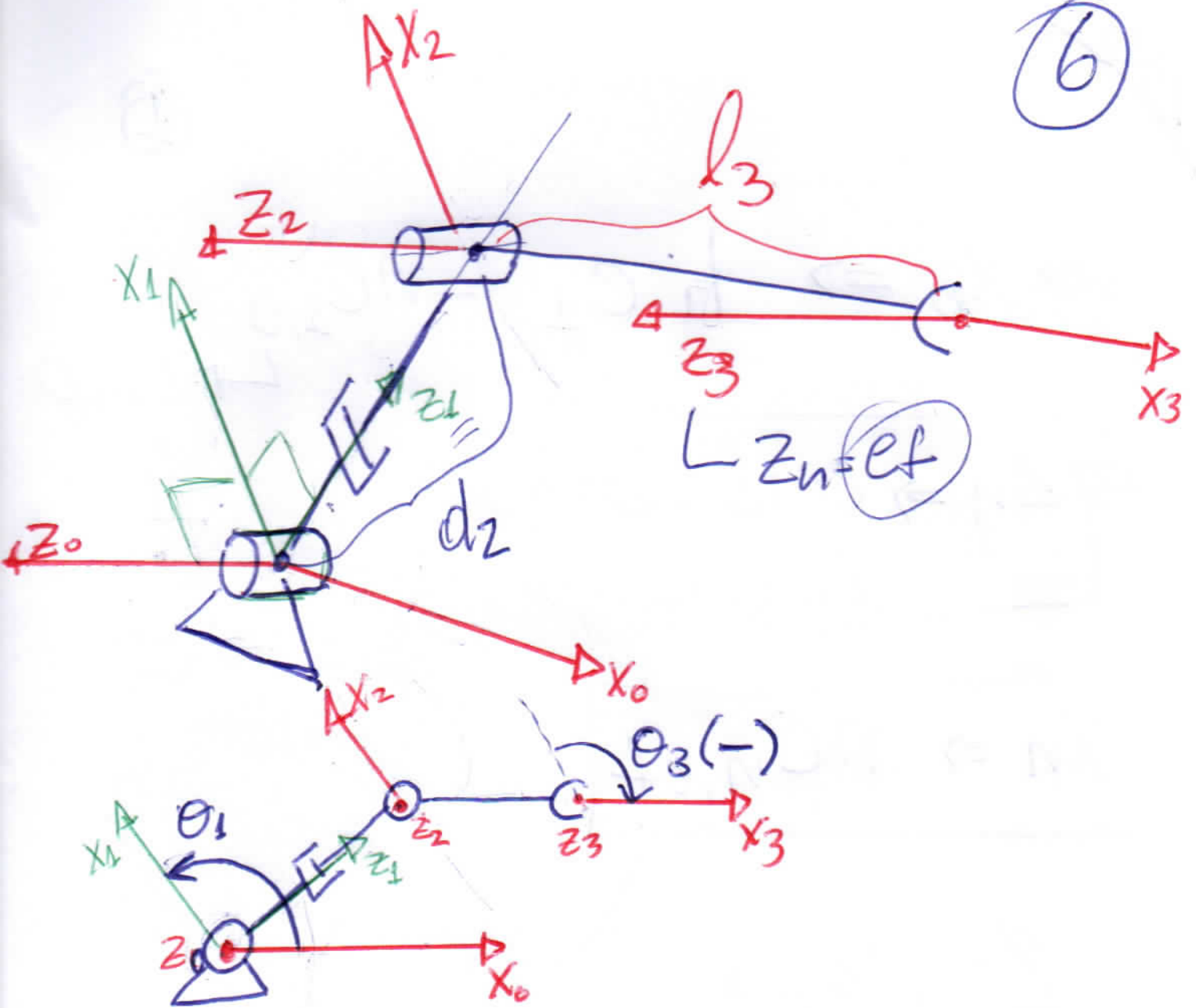
Para juntas prismáticas



$\theta_i =$ fixo

$d_i =$ variável

⑥



i	a_i	L_i	θ_i	d_i	
1	0	$\pi/2$	θ_1	0	R
2	0	$-\pi/2$	0	d_2	P
3	l_3	0	θ_3	0	R

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(7)

$$a_1 \Rightarrow n_{C12} = 0$$

Lsegmento

$$\alpha_1 \Rightarrow z_0 \xrightarrow{x_1} z_1 = \pi/2$$

$$\theta_1 \Rightarrow x_0 \xrightarrow{z_0} x_1 = \theta_1$$

$$d_1 \Rightarrow \checkmark$$

$$a_3 = n_{C3e}$$

Ll3

$$\alpha_3 = z_2 \xrightarrow{x_3} z_3$$

$$\hookrightarrow 0 \text{ rad}$$

$$\theta_3 \Rightarrow x_2 \xrightarrow{z_2} x_3$$

$$d_3 \Rightarrow$$

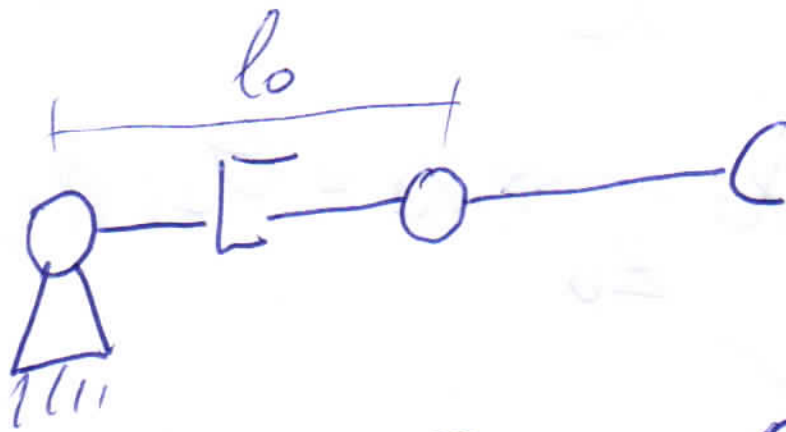
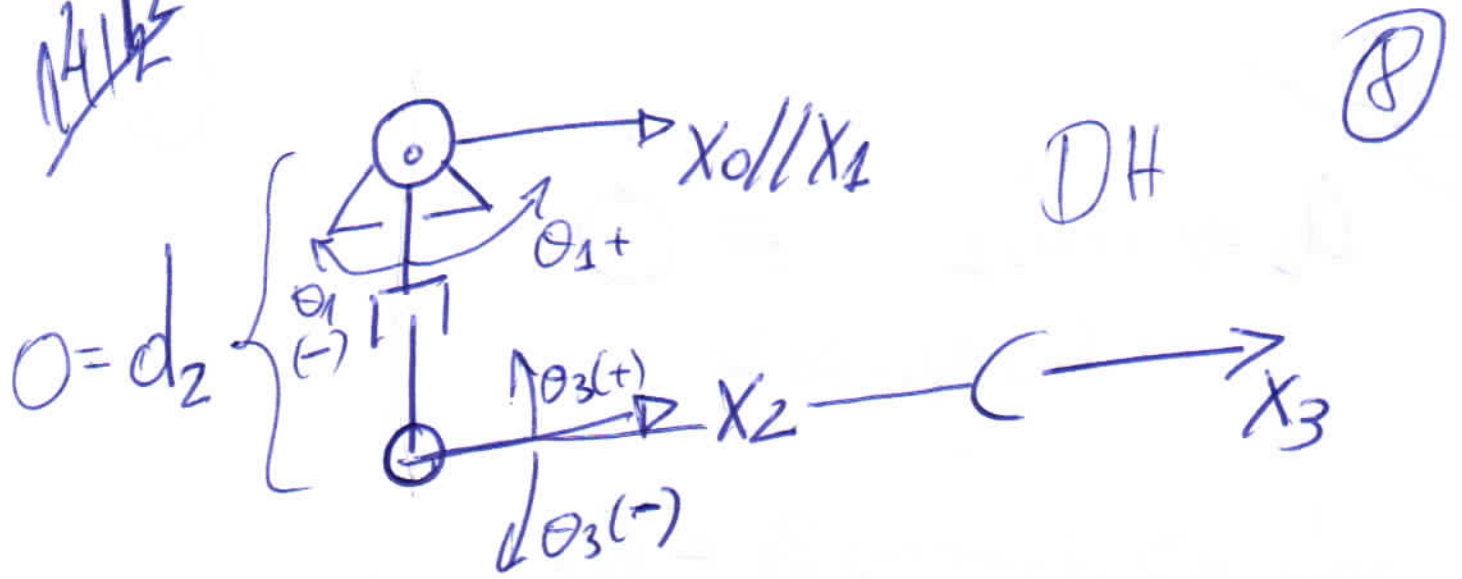
$$a_2 \Rightarrow n_{23} = 0$$

$$\alpha_2 \Rightarrow z_1 \xrightarrow{x_2} z_2 = -\pi/2$$

$$\theta_2 \Rightarrow x_1 \xrightarrow{z_1} x_2 = 0$$

$$d_2 \Rightarrow \checkmark$$

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$$\theta_{10} = +\pi/2 \Rightarrow \theta_1 = \theta_{1r} + \pi/2$$

$$d_{20} = l_0 \Rightarrow d_2 = d_r + l_0$$

$$\theta_{30} = -\pi/2 \Rightarrow \theta_3 = \theta_{3r} - \pi/2$$

UPSCV3SS

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$$A_1^0 = \begin{bmatrix} C_1 & 0 & S_1 & 0 \\ S_1 & 0 & -C_1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2^1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_3^2 = \begin{bmatrix} C_3 & S_3 & 0 & l_3 C_3 \\ S_3 & C_3 & 0 & l_3 S_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_3^0 = A_1^0 A_2^1 A_3^2$$

$$T_6^0 = \left[\begin{array}{c|c} R_6^0 & P_6^0 \\ \hline 000 & 1 \end{array} \right] = \underset{\uparrow}{A_1} A_2^1 A_3^2 A_4^3 A_5^4 A_6^5$$

	a	2	0	d
1	0	$\pi/2$	θ_1	L_1
2	L_2	0	θ_2	0
3	0	$\pi/2$	θ_3	0
4	0	$-\pi/2$	θ_4	L_4
5	0	$\pi/2$	θ_5	0
6	0	$-\pi/2$	θ_6	L_6